Review of Chapter 1.

1. First-order differential equations (general form)

$$\frac{dy}{dx} = f(x,y)$$

Initial value problem

$$\begin{cases} \frac{dy}{dx} = f(x,y) \\ y(x_0) = y_0 \end{cases}$$

2. Simple form of differential equation.

$$\frac{dy}{dx} = f(x).$$

then $$y(x) = \int f(x)\,dx + C$$
$$= G(x) + C$$

$$\begin{cases} \frac{dy}{dx} = f(x) \\ y(x_0) = y_0 \end{cases} \Rightarrow y(x) = G(x) + y_0 - G(x_0)$$

or

$$y(x) = \int_{x_0}^{x} f(t)\,dt + y_0$$

3. Second-order equation.

$$\frac{d^2y}{dx^2} = g(x). \Rightarrow \begin{cases} \frac{dy}{dx} = V(x) \\ \frac{dV}{dx} = g(x) \end{cases} \begin{array}{l} \Rightarrow y(x) = \int G(x)\,dx + C_1 x + C_2 \\ V(x) = \int g(x)\,dx = G(x) + C_1 \end{array}$$

Particle motion: $X(t) = X(t_0) + \int_{t_0}^{t} V(s)\,ds$

Constant Acceleration.

$$\begin{cases} V(t) = V_0 + at \\ X(t) = \frac{1}{2}at^2 + V_0 t + X_0 \end{cases}$$

4. Existen and Uniqueness for the solution of

$$\frac{dy}{dx} = f(x,y), \quad y(a) = b.$$

· If $f(a,b)$ is continuous at (a,b), then the solution exists near $x=a$.
· If $\frac{\partial f}{\partial y}(a,b)$ is also continuous at (a,b), then the solution is unique near $x=a$.

5. Separable equation.

$$\frac{dy}{dx} = \frac{g(x)}{f(y)}$$

then $$\int f(y)\,dy = \int g(x)\,dx.$$

$$F(y) = G(x) + C.$$

Remark: One may not be able to express y explicitly in term of x. then we just need write down the implicit solution.

$$H(x,y) = F(y) - G(x) = C$$

$$\begin{cases} \frac{dy}{dx} = \frac{g(x)}{f(y)} \\ y(x_0) = y_0 \end{cases} \Rightarrow F(y) - G(x) = F(y_0) - G(x_0)$$

6. Natural Growth and Decay.

$$\frac{dx}{dt} = kx \qquad k - \text{Constant} \ \left(\begin{matrix} k>0 & - \text{Growth} \\ k<0 & - \text{decay} \end{matrix} \right)$$

population Growth: $\frac{dP}{dt} = (\underbrace{\beta}_{\text{Birth rate}} - \underbrace{\alpha}_{\text{death rate}})P \rightarrow P = P_0 e^{(\beta-\alpha)t}$

Compound interest: $\frac{dA}{dt} = \underbrace{r}_{\text{compound interest rate}} A \rightarrow A = A_0 e^{rt}$

Radioactive Decay: $\frac{dN}{dt} = -\underbrace{k}_{\text{decay rate}} N \rightarrow N = N_0 e^{-kt}$

For ^{14}C, we have $k = 0.0001216$ if t is measured by year.

cooling and Heating: $\frac{dT}{dt} = k(A-T)$ $\qquad$ A: Environmental temperature.

$$\rightarrow T(t) = A - (A - T_0)e^{-kt}$$

7. Linear first order differential equation

$$\frac{dy}{dt} + P(x)y = Q(x)$$

Case I: $Q(x) \equiv 0$, then $y(x) = Ce^{-\int P(x)dx}$

Case II: $Q(x) \neq 0$, then.

Integrating factor: $\rho(x) = e^{\int P(x)dx}$

$$\frac{d}{dx}[\rho(x)y] = \rho(x)Q(x)$$

$$\Rightarrow \rho(x)y = \int \rho(x)Q(x) + C$$

$$y = \frac{1}{\rho(x)}\left(\int \rho(x)Q(x) + C\right)$$ → General solution.

Initial value problem

$$\begin{cases} \frac{dy}{dx} + P(x)y = Q(x) \\ y(x_0) = y_0 \end{cases}$$

Two ways of solution.

i) Find the general solution, then determine the constant C using initial values.

ii) Choose the integrating factor

$$\rho(x) = e^{\int_{x_0}^{x} P(t)dt}$$

then

$$y(x) = \frac{1}{\rho(x)}\left[\int_{x_0}^{x} \rho(\,)Q(t)dt + y_0\right]$$

8. Mixture problem:

$$\frac{dx}{dt} = r_i C_i - r_o C_o(t)$$
$$= r_i C_i - \frac{r_o}{V(t)}x(t)$$
$$= r_i C_i - \frac{r_o}{V_0 + (r_i - r_o)t}x$$

r_i : inflow rate.
C_i : inflow concentration
r_o : outflow rate
C_o : outflow concentration
V : Volume of the solution
x : the amount of solute.

9. Substitution methods

$$\frac{dy}{dx} = f(x,y) \quad \to \text{General situation.}$$

let $V=\alpha(x,y)$, $y=\beta(x,V)$, $\frac{dy}{dx} = \frac{\partial\beta}{\partial x} + \frac{\partial\beta}{\partial V}\frac{dV}{dx}$, then we have

$$\frac{dV}{dx} = \frac{f(x,\beta(x,V)) - \beta_x}{\beta_V} = g(x,V) \to \begin{matrix}\text{seperable ?} \\ \text{linear ?}\end{matrix}$$

i) Homogeneous Equation.

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

let $\underline{V=\frac{y}{x}}$, $y=Vx$, $\frac{dy}{dx} = V + x\frac{dV}{dx}$, then

$$V + x\frac{dV}{dx} = F(V) \Rightarrow \frac{dV}{dx} = \frac{F(V)-V}{x} \quad \to \text{seperable.}$$

ii) Bernoulli equation.

$$\frac{dy}{dx} + P(x)y = Qy^n \qquad (n=0,1 \to \text{linear}).$$

Let $\underline{V=y^{1-n}}$, $y=V^{\frac{1}{1-n}}$, $\frac{dy}{dx} = \frac{1}{1-n}V^{\frac{n}{1-n}}\frac{dV}{dx}$, then

$$\frac{1}{1-n}V^{\frac{n}{1-n}}\frac{dV}{dx} + P(x)^{\frac{1}{1-n}} = Q(x)V^{\frac{n}{1-n}}$$

$$\Rightarrow \frac{dV}{dx} + (1-n)P(x)V = (1-n)Q(x) \quad \to \text{linear.}$$

10. Consider the equation.

$$M(x,y)dx + N(x,y)dy = 0$$

If $\frac{\partial M}{\partial y}(x,y) \equiv \frac{\partial N}{\partial x}(x,y) \Longrightarrow$ This equation is exact!

Method for finding solution. $F(x,y)=C$

1. $F(x,y) = \int M(x,y)dx + g(y) \quad \left(\frac{\partial F}{\partial x} = M\right)$
2. Compute $g'(y)$ by using $\frac{\partial F}{\partial y} = N(x,y)$

$$\Rightarrow g'(y) = \int \left[N - \frac{\partial}{\partial y}\left(\int M(x,y)dx\right)\right]dy$$

3. Then $F(x,y) = \int M(x,y)dx + \int \left[N - \frac{\partial}{\partial y}\left(\int M(x,y)dx\right)\right]dy$